

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12387893
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Delbaere; Stéphane et al.

Method for diagnosing an operating state of an electrical switching device and electrical switching device for implementing such a method

Abstract

A method for diagnosing an operating state of a switching device including separable contacts, which are driven by an electromagnetic actuator including a coil connected to an electronic control device, and sensors, which are configured to measure an intensity of a coil current and a coil voltage. The method comprises: a) receiving an order to close via the electronic control device; b) controlling the electromagnetic actuator so that it closes; c) measuring and recording the coil voltage and coil current values; d) calculating and recording values of a magnetic flux through the coil, by integrating the recorded values of the coil current, the coil voltage and resistance and inductance values of the coil; and e) evaluating and recording positions of a core of the electromagnetic actuator according to a data table, which is previously recorded in the electronic control device and which defines a bijective relation between the position of the core, the magnetic flux and the coil current.

Inventors: Delbaere; Stéphane (Meylan, FR), Orban; Rémy (Saint Martin d'Uriage, FR)

Applicant: Schneider Electric Industries SAS (Rueil Malmaison, FR)

Family ID: 73013613

Assignee: Schneider Electric Industries SAS (Rueil Malmaison, FR)

Appl. No.: 17/376248

Filed: July 15, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220020546 A1	Jan. 20, 2022

Foreign Application Priority Data

FR	2007585	Jul. 20, 2020
----	---------	---------------

Publication Classification

Int. Cl.: H01H47/00 (20060101); G01R31/327 (20060101); H01H1/00 (20060101)

U.S. Cl.:

CPC H01H47/002 (20130101); G01R31/327 (20130101); G01R31/3274 (20130101); H01H1/0015 (20130101);

Field of Classification Search

CPC: G01R (31/327); G01R (31/3271); G01R (31/3272); G01R (31/3275); G01R (31/3274); G01R (31/3277); G01R (31/3278); H01H (1/0015); H01H (47/002)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4025886	12/1976	Barkan	335/147	H01H 1/54
5424637	12/1994	Oudyn	324/207.16	H01F 7/1844
6208497	12/2000	Seale	361/160	H02K 41/03
6518748	12/2002	Butzmann	324/207.16	F01L 9/20
6657847	12/2002	Wright	361/160	H01F 7/1844
6859350	12/2004	Corbetta	361/139	H01H 33/593
7109720	12/2005	Baurand et al.	N/A	N/A
8688391	12/2013	Elsner	702/34	H01H 1/0015
9733292	12/2016	Delbaere	N/A	G06F 15/00
11948757	12/2023	Rushabh	N/A	G01R 31/3274
2017/0199231	12/2016	Brunner et al.	N/A	N/A
2019/0018065	12/2018	Lapiere	N/A	H01H 11/0062
2019/0131047	12/2018	Priest	N/A	H01F 7/064
2019/0348243	12/2018	Chen	N/A	G01R 31/327
2020/0194191	12/2019	Ashtekar	N/A	G01R 31/3333

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3460822	12/2018	EP	N/A
9824106	12/1997	WO	N/A
03054895	12/2002	WO	N/A
2015189027	12/2014	WO	N/A

OTHER PUBLICATIONS

French Search Report and Written Opinion dated Mar. 11, 2021 for French Patent Application No. 2007585, 7 pages. cited by applicant

Primary Examiner: Nguyen; Judy

Assistant Examiner: Split; James

Attorney, Agent or Firm: Troutman Pepper Locke LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a method for diagnosing an operating state of an electrical switching apparatus and to electrical switching apparatus for implementing such a method.

(2) The invention more particularly relates to electrical contactors.

BACKGROUND

(3) Such electrical switching apparatuses comprise an electromagnetic actuator comprising a coil and are configured to switch between an open state and a closed state, for example in order to control the supply of electrical power to an electrical load. Generally, the electrical contacts comprised fixed contacts and movable contacts, the latter being attached to a movable portion of the actuator that moves under the effect of a magnetic field created by the coil when a suitable electrical current flows through the latter. To guarantee a sufficient contact pressure between the electrical contacts, the movable portion of the actuator overruns by a distance, called the overrun, between the moment at which the movable contacts make contact with the fixed contacts, and the moment at which the actuator reaches its stable closed position. This overrun corresponds to a contact compression.

(4) In each switching cycle, the contactor undergoes wear under the effect of various factors; for example, the electrical contacts wear under the effect of electric arcs generated as the electrical contacts open, the wear of the contactor manifesting itself by a loss of contact compression.

(5) It would be desirable to be able to automatically estimate the wear level of a contactor while it is in operation, in order to be able to make provision for suitable maintenance during the life of said contactor and/or to detect the appearance of malfunctions.

(6) It is known to incorporate a position sensor into the electromagnetic actuator, in order to directly measure the movement of the movable parts of the actuator and to deduce contact compression therefrom. However, the use of a sensor engenders an additional cost, and it is not always possible to integrate a sensor into a contactor.

(7) WO-03/054895-A1 describes a method for diagnosing wear based on a measurement of the travel time of the movable core of the electromagnetic actuator in the closing phase. Such a diagnosing method only allows the time of the start of the movement and the time of the end of the movement of the core to be determined. Thus, one drawback of this method is that it does not allow the precise position of the core during the closing phase to be determined. This makes it more difficult to obtain reliable information on the state of the contacts and therefore on the wear state of the contactor.

SUMMARY

(8) It is these problems that the invention more particularly remedies, by providing a diagnosing method that is more precise.

(9) To this end, the invention relates to a method for diagnosing an operating state of a switching apparatus that is configured to be coupled to an electrical conductor and that comprises: separable contacts that are associated with the electrical conductor and that are driven by an electromagnetic actuator comprising a coil connected to an electronic control device, which control device is configured to apply a coil command voltage across the terminals of the coil, and sensors that are

configured to measure a magnitude of a coil current flowing through the coil and the coil voltage. According to the invention, the method comprises steps consisting in: receiving an order to close the switching apparatus, the switching apparatus initially being in an open state, the order to close being received by the electronic control device; after the order to close has been received, commanding the electromagnetic actuator to close, by means of the electronic control device; as the switching apparatus switches to the closed state, measuring and storing values of the coil voltage and coil current; computing and storing values of a magnetic flux passing through the coil, by integration of the stored values of the coil current and coil voltage and of values of the resistance and inductance of the coil that are stored beforehand in the electronic control device; on the basis of the stored values of the magnetic flux and coil current, evaluating and storing positions of a core of the electromagnetic actuator depending on a data table characteristic of the electromagnetic actuator, the data table being stored beforehand in the electronic control device and defining a bijective relationship between the position of the core, the magnetic flux and the coil current.

(10) By virtue of the invention, the magnetic flux is computed throughout the closing phase on the basis of the measurements of coil voltage and of coil current. A value of the position of the movable core is deduced therefrom on the basis of the data table, which relates bijectively the position of the core, coil flux and coil current. The variation in the position of the mobile core in the closing phase is thus determined automatically, precisely, and without having to install a specific new sensor such as a position sensor, this being economical.

(11) According to some advantageous but non-mandatory aspects of the invention, such a diagnosing method may incorporate one or more of the following features, either alone or in any technically permissible combination: the step of computing the magnetic flux comprises an initial substep, referred to as the self-correction substep, followed by a computing substep, the self-correction substep consisting in, provided that the coil current is lower than a current referred to as the “start” current, below which the movable core is kept in the open position by a return member of the electromagnetic actuator, evaluating and storing an initial value of the magnetic flux, which value is referred to as the “initial flux”, so that, when the coil current increases above the start current, the computation by integration of the magnetic flux in the computing phase takes into account the value of the initial flux; in the step of commanding the electromagnetic actuator to close, the electronic control device commands the electromagnetic actuator to close when the coil current reaches 50% of the value of the start current, preferably, and more preferably the start current. the method comprises a step consisting in computing, from the profile of the position, which profile is stored in the electronic control device, a closing time of the electromagnetic actuator, the closing time being equal to a time difference between the time at which the electronic control device commands the electromagnetic actuator to close and the time at which the electromagnetic actuator reaches its stable closed position; the method comprises a step consisting in computing, by derivation with respect to time of the profile of the position stored in the electronic control device, a profile of the speed of movement of the switching apparatus and a profile of the acceleration of the switching apparatus, and in storing the speed and acceleration profiles in the electronic control device; the method comprises a step consisting in computing, from the speed profile of the switching apparatus, a minimum compression speed of the core, the minimum compression speed being equal to a minimum of the speed of the core in a terminal segment of the movement of the core in the closing phase; the method detects the precise times of closure of each contact associated with each pole of the switching apparatus by means of current sensors, the method combining the speed profile stored in the electronic control device and the precise times of closure of each contact to evaluate a closing speed of each contact; and a compression associated with the movable contacts of each pole is evaluated by combining the position profile of the core with the precise times of closure of each contact, the compression of each movable contact being equal to the difference between a final position of the core in its stable

closed position and the position of the core at the precise time of closure of the contact in question.

(12) The invention also relates to an electrical switching apparatus for implementing a diagnosing method such as described above, the switching apparatus being configured to be coupled to an electrical conductor and comprising: separable contacts, which are moved between an open position and a closed position by an electromagnetic actuator comprising a coil and a movable core that is attached to the separable contacts, the switching apparatus having a structure that limits the generation of eddy currents; a command circuit for controlling the voltage across the terminals of the coil, which voltage is referred to as the “coil voltage”, the command circuit comprising a device referred to as the “release” device, which may be selectively activated in order to make drop the electrical current flowing through the coil, which is referred to as the “coil current”, the coil voltage and the release device being activated or deactivated depending on the states of the command circuit; sensors for measuring the coil current and coil voltage; an electronic control device, which is configured to receive orders to open and close the switching apparatus, to receive the values of the measurements of coil current and coil voltage, and to control the states of the command circuit; wherein the switching apparatus is configured to implement a diagnosing method comprising steps consisting in: a) receiving an order to close; b) commanding the electromagnetic actuator to close; c) measuring and storing values of the coil voltage and coil current; d) computing and storing values of a magnetic flux passing through the coil, by integration of the stored values of the coil current and coil voltage and of values, which are stored beforehand in the electronic control device, of the resistance and inductance of the coil; e) on the basis of the stored values of the magnetic flux and coil current, evaluating and storing positions of the core depending on a data table characteristic of the electromagnetic actuator, the data table defining a bijective relationship between the position of the core, the magnetic flux and the coil current.

(13) Advantageously, the switching apparatus comprises current sensors, which are configured to measure a magnitude of the electrical current flowing through each phase of the electrical conductor, the switching apparatus being configured to implement a diagnosing method comprising steps consisting in detecting a precise time of closure of the movable contacts associated with each phase, and in combining with the precise times of closure of each of the movable contacts the position profile of the core or the speed profile of the core, in order to evaluate a compression or a closing speed of each of the movable contacts.

(14) This switching apparatus has the same advantages as those mentioned above with respect to the diagnosing method of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will be better understood and other advantages thereof will become more clearly apparent from the following description of one embodiment of a method for diagnosing an operating state of a contactor and of a contactor configured to implement such a method, which description is given solely by way of example and with reference to the appended drawings, in which:
- (2) FIG. 1 is a schematic representation of an electric switching apparatus comprising an electromagnetic actuator, according to embodiments of the invention;
- (3) FIG. 2 is a schematic view of an example of the command circuit for controlling the electromagnetic actuator of the switching apparatus of FIG. 1;
- (4) FIG. 3 is a graph showing the variation in an electrical command current of the electromagnetic actuator of FIG. 2, in a plurality of operating phases;
- (5) FIG. 4 is a graph showing the variation in an electrical command current and in a magnetic flux of the electromagnetic actuator of FIG. 2, as measured in a phase of closing the actuator;

- (6) FIG. 5 is a graph illustrating one of the steps of the diagnosing method;
(7) FIG. 6 is a graph illustrating a result of the diagnosing method; and
(8) FIG. 7 is a chart showing steps of a method for diagnosing properties of the switching apparatus of FIG. 1, according to embodiments of the invention.

DETAILED DESCRIPTION

- (9) A contactor **1** is shown in FIG. 1. The contactor **1** is an example of an electrical switching apparatus intended to control the supply of electrical power from an electrical power source to an electrical load **23**. The electrical power source has not been illustrated to simplify the figures. The power source is for example a mains grid, whereas the electrical load **23** is for example an electric motor that it is desired to control and/or protect by means of the contactor **1**. The contactor **1** is generally housed in a casing, which has been represented here by a dashed rectangle. The contactor **1** is configured to be coupled, on the one hand, to an upstream electrical conductor **20** connected to an electrical power source and, on the other hand, to a downstream line **22** connected to the electrical load **23**. When the contactor **1** lets electrical current pass and the electrical load **23** is supplied with power, the contactor **1** is in a state referred to as the “closed” state, whereas when the contactor **1** prevents the passage of electrical power to the electrical load, the contactor **1** is in a state referred to as the “open” state.
- (10) The electrical conductor **20** and the downstream line **22** comprise the same number of phases. When the electrical power source is multiphase, the electrical conductors **20** and the downstream line **22**, which each comprise conductive wires that are insulated from one another, have as many conductive wires as each other, each conductive wire of the downstream line **22** being associated with one respective wire of the upstream electrical conductor **20**. Whatever the number of phases, the contactor **1** is configured to interrupt, or to let pass, the electrical current conjointly in each of the phases.
- (11) In the illustrated example, the upstream electrical conductor **20** is three-phase. A single of the conductive wires of the electrical conductor **20**, which has been referenced **201**, is shown in FIG. 1. Only the conductive wire of the power line **22** associated with the conductive **201** has been shown, this conductive wire of the power line **22** being referenced **221**.
- (12) The rest of the description will be given with reference to the conductive wires **201** and **221**, which are associated with the same phase of the supply current, but it will be understood that what is described may be transposed to the other phases of the supply current.
- (13) For each phase, the contactor **1** comprises movable contacts **24**, which are placed on a movable bar **26**, and fixed contacts **28**, which are attached to the upstream and downstream conductors **20**, **22**, respectively. Each of the movable and fixed contacts **24**, **28** comprises contact pads **29**, which are here made of metal, and preferably from a silver alloy or any equivalent material.
- (14) The movable bar **26** is movable between a closed position, in which the movable contacts **24** are electrically connected to the fixed contacts **28** and electrical power is able to flow through the movable bar **26** of the upstream electrical conductor **20** to the downstream line **22**, and an open position, in which the removable contacts **24** are separated from the fixed contacts **28**.
- (15) When the movable bar **26** is in closed position, the contactor **1** is in the closed state, whereas when the bar **26** is in open position, the contactor **1** is in the open state. The passage from the open state to the closed state is a phase of closing of the contactor **1**, whereas the passage from the closed state to the open state is a phase of opening of the contactor **1**.
- (16) In practice, in each cycle comprising a closing and opening phase, the contact pads **29** wear, for example under the action of electric arcs generated as the contactor opens, or even because of material being torn away as a result of micro-welds. The consequence of this loss of material is that the thickness of the contact pads **29** decreases throughout the life of the contactor **1**, this increasing the amplitude of the movement of the bar **26** during opening and closing phases. To remedy this, the contactor **1** comprises a mechanism **290**, which has been schematically represented by a spring

in FIG. 1, that is attached to the bar **26** and that allows the fixed and movable contacts **28**, **24** to be kept in electrical contact with a sufficient contact pressure.

(17) When the thickness of the contact pads **29** is insufficient or when the surface finish of the pads **29** is poor, the risk of malfunction of the contactor **1** increases and it is recommendable to replace the contactor **1**; it is for these reasons that a diagnosis of the compression state of the contacts of the contactor **1** allows the advance of the degradation of the contactor **1** to be evaluated.

(18) The movable bar **26** is driven by an electromagnetic actuator **30**, which comprises a command electromagnet with a coil **32**, a core **34** attached to the movable bar **26** and a return member **36**, a spring or equivalent for example. The coil **32** is configured to generate a magnetic field when it is powered with an electrical command current, in order to cause the core **34**, and therefore the movable bar **26**, to move. The movements of the core **34** between the open and closed positions have been represented by a double-headed arrow **F34**. In other words, the movable contacts **24** and the associated fixed contacts **28** together form separable contacts, which are associated with the electrical conductor **20** and which are moved between an open position and a closed position by the electromagnetic actuator **30**, which comprises a coil **32** and a movable core **34** that is attached to the separable contacts.

(19) In FIG. 1, the contactor **1** has been shown in an intermediate configuration between the stable open and closed states of the contactor **1**, in which intermediate configuration the fixed and movable contacts **28**, **24** are electrically connected, but the core **34** is not in abutment in the closed position. The mechanism **290** permits an overrun of the core **34** between the moment at which the fixed contacts **28** make contact with the movable contacts **24**, and the moment at which the actuator **30** is in its stable closed position. This overrun, which corresponds to the contact compression, is denoted **E** in FIG. 1.

(20) The electromagnetic actuator **30** is controlled by way of a supply circuit **38**, itself controlled by an electronic control device **40**. The coil **32** is thus connected to the electronic control device **40**.

(21) According to some embodiments, the electronic control device **40** comprises a central processing unit (CPU), such as a programmable microcontroller, a microprocessor or the like, and a computer memory forming a storage medium for computer-readable data.

(22) According to some examples, the memory is a ROM memory, a RAM memory, or an EEPROM or flash non-volatile memory or the like. The memory comprises executable instructions and/or computer code for ensuring the operation of the control device **40** in accordance with one or more of the embodiments described below when executed by the central processing unit.

(23) As variants, the electronic control device **40** may comprise a digital signal processor (DSP), or a field-programmable gate array (FPGA), or an application-specific integrated circuit (ASIC), or any equivalent element.

(24) The electronic control device **40** is itself connected to a power supply rail **42** and comprises an interface **44** configured to receive, from a user, orders to open or close the contactor **1**. The electronic control device **40** is here shown as being integrated into the contactor **1**. As a variant, the control device **40** is remote, i.e. it is not integrated into the same casing as the electromagnetic actuator **30**.

(25) The power supply rail **42** has a preferably stable and DC voltage and is intended to supply the electronic control device **40** and the supply circuit **38** with power. The interface **44** has been represented here by command electrodes. For example, an electrical command voltage may be applied across the command electrodes. Optionally, the interface **44** comprises wireless communication means.

(26) In certain embodiments, the contactor **1** also comprises current sensors **46**, configured to measure a current flowing through each of the phases of the upstream line **20**, or in other words a current flowing through each of the conductive wires **201** of the upstream line **20**. In other embodiments, the current sensors **46** and the electronic control device **40** are integrated into a casing separate from the contactor **1**.

(27) When the coil **32** is supplied with electrical power delivered by the power supply rail **42**, an excitation current passes through the coil **32**, which generates an electromagnetic force that tends to attract the core **34** and the bar **26** from the open position to the closed position. A return member **36**, represented here by a spring, exerts a return force that opposes the force of attraction of the electromagnet.

(28) A coil current $I_{\text{sub.BOB}}$ is defined as being an excitation current flowing through the coil **32**.

(29) A start current $I_{\text{sub.D}}$ is defined as being a threshold of the coil current $I_{\text{sub.BOB}}$ that, when the actuator **1** is in the open state, allows the actuator **1** to move to the closed state, as soon as the coil current $I_{\text{sub.BOB}}$ increases above the start current $I_{\text{sub.D}}$.

(30) A stall current $I_{\text{sub.S}}$ is defined as being a threshold of the coil current $I_{\text{sub.BOB}}$ that, when the actuator **1** is in the closed state, allows the actuator **1** to move to the open state, as soon as the coil current $I_{\text{sub.BOB}}$ decreases below the stall current $I_{\text{sub.S}}$.

(31) Thus, when that the actuator **1** is in the open state, provided that the coil current $I_{\text{sub.BOB}}$ remains below the start current $I_{\text{sub.D}}$, the movable core **34** is repulsed to the open position by the return member **36** of the actuator **30** and the contactor **1** remains in the open state. If the coil current $I_{\text{sub.BOB}}$ increases above the start current $I_{\text{sub.D}}$, the electromagnetic force of the coil **32** becomes higher than the return force of the member **36** and the core **34** then moves from its open position to its closed position. Such a situation corresponds to a phase of closing the contactor **1**.

(32) In contrast, when the actuator **1** is in the closed state, provided that the coil current $I_{\text{sub.BOB}}$ remains higher than the stall current $I_{\text{sub.S}}$, the contactor **1** remains in the closed state. If, while the contactor **1** is in the closed state, the coil current $I_{\text{sub.BOB}}$ decreases below the stall current $I_{\text{sub.S}}$, the electromagnetic force of the coil **32** drops below the return force of the member **36** and of the mechanism **290**; the core **34** is then repulsed from its closed position to its open position under the effect of the return member **36** and of the mechanism **290**. Such a situation corresponds to a phase of opening the contactor **1**.

(33) Generally, the start current $I_{\text{sub.D}}$ is higher, in absolute value, than the stall current $I_{\text{sub.S}}$. The values of the start and stall currents $I_{\text{sub.D}}$, $I_{\text{sub.S}}$ are adjusted during the design of the actuator **1**, especially by adjusting the properties of the coil **32** or the return forces of the member **36** and of the mechanism **290**.

(34) An exemplary embodiment of the supply circuit **38** is schematically shown in FIG. 2. The architecture of the supply circuit **38** is nonlimiting, and other arrangements of the various components of the supply circuit **38** are possible, and indeed other electrical or electronic components may be used to perform the same functions.

(35) The supply circuit **38** preferably comprises a measurement circuit **50**, shown in the left-hand dashed rectangle, configured to measure the value of the electrical voltage between the supply rail **42** and an electrical ground GND of the supply circuit **38**.

(36) For example, the measurement circuit **50** comprises two resistors **R1** and **R2** that are connected in series with a diode **Dt** between the power supply rail **42** and the electrical ground GND. A first measurement point, here placed between the resistors **R1** and **R2**, allows a first measurement voltage **V1** representative of the voltage present between the power supply rail **42** and the electrical ground GND to be collected. Since the electrical voltage present between the rail **42** and the ground GND is related to the coil voltage $U_{\text{sub.BOB}}$, the measurement circuit **50** is by extension an example of a sensor configured to measure coil voltage $U_{\text{sub.BOB}}$.

(37) The supply circuit **38** comprises a command circuit **51**, which includes the coil **32**. A diode **D1** may be placed on the power supply rail **42** between the command circuit **51** and the measurement circuit **50** in order to prevent any return of current to the latter. The diodes **D1** and **Dt** are preferably of the same type.

(38) The command circuit **51** is here shown in a configuration referred to as the “release” configuration, which is described in more detail further on in the present description.

(39) The command circuit **51** comprises a supply terminal **52**, which is connected to the power

supply rail **42**. The coil **32** comprises two terminals **54** and **56**. A measurement of the voltage between the terminals **54** and **56** thus allows a coil voltage, which is denoted $U_{\text{sub.BOB}}$, to be measured.

(40) The terminal **54** is connected to ground GND via a switch **t1** referred to as the “release” switch. In many embodiments, a resistor R_{sh} , referred to as the shunt resistor, is connected in series with the release switch **T1** in order to collect a second measurement voltage V_2 representative of the electrical current flowing through the coil **32**, or in other words representative of the coil current $I_{\text{sub.BOB}}$. In the illustrated example the shunt resistor R_{sh} is connected between the release switch **T1** and ground GND. The resistor R_{sh} is an example of a sensor configured to measure the magnitude of the coil current $I_{\text{sub.BOB}}$ flowing through the coil **32**.

(41) The terminal **56** is, on the one hand, connected to the terminal **52** by way of a supply switch **T2** and, on the other hand, connected to ground GND by way of a diode, which is referred to as the “flyback” diode D_{rl} . The flyback diode D_{rl} has a blocking direction oriented toward the terminal **56**.

(42) The switches **T1** and **T2** are switches that are controlled by a command signal originating from the electronic control device **40**. In other words, the electronic control device **40** is configured to control the states of the command circuit **51**.

(43) According to some examples of implementation, the switches **T1** and **T2** are semiconductor power switches, such as MOSFETs, or thyristors, or insulated-gate bipolar transistors (IGBT), or any other equivalent devices.

(44) The command circuit **51** comprises a “release” device D_z , which here takes the form of a Zener diode connected in parallel to the release switch **T1**. Thus, when the release switch **T1** is opened, the coil current $I_{\text{sub.BOB}}$ passes through the release device D_z , whereas when the release switch **T1** is closed, the release device D_z is short-circuited and no current passes through the release device D_z . The release device D_z is thus selectively activatable to make the coil current $I_{\text{sub.BOB}}$ drop.

(45) In the context of the present invention, the characteristic quantities of the components of the supply circuit **38** are considered to be known. In particular, the coil **32** has a resistance denoted $R_{\text{sub.BOB}}$ and an inductance denoted $L_{\text{sub.BOB}}$. The resistance $R_{\text{sub.BOB}}$ and inductance $L_{\text{sub.BOB}}$ especially depend on the geometry of the coil **32**, on the materials used, on temperature, etc. The values of the resistance $R_{\text{sub.BOB}}$ and inductance $L_{\text{sub.BOB}}$ are thus considered to be known.

(46) FIG. 3 shows a graph **58** illustrating the variation in the coil current $I_{\text{sub.BOB}}$ flowing through the coil **32** over time (t) in various successive operating phases of the contactor **1**, which phases are denoted **P1**, **P2**, **P3** and **P4**, in the case where the contactor is switched to the closed state, then switched back to the open state.

(47) The phase **P1** is an initial phase, during which the contactor **1** is stably in the open state, i.e. the coil current $I_{\text{sub.BOB}}$ does not exceed the start current $I_{\text{sub.D}}$. In FIG. 3, the coil current $I_{\text{sub.BOB}}$ contains transient peaks **60** and **62**, which are lower than the start current $I_{\text{sub.D}}$ and which are related to operations for testing the contactor **1**, these testing operations not being described in detail in the present description.

(48) The phase **P2** corresponds to a closing phase, after an order to close has been received by the contactor **1**. The phase **P2** starts when the electronic control device **40** receives, at a time $t_{\text{sub.0}}$, an order to close via the interface **44**.

(49) The electronic control device **40** is configured to apply the command coil voltage $U_{\text{sub.BOB}}$ to the terminals of the coil **32**. For example, the control device **40** then commands the release switch **T1** and the supply switch **T2** to close. The coil voltage $U_{\text{sub.BOB}}$ is then equal to the voltage delivered by the power supply rail **42**, minus the voltage across the terminals of the shunt resistor R_{sh} . The coil current $I_{\text{sub.BOB}}$, which is initially zero, then increases to exceed the start current $I_{\text{sub.D}}$ at a time $t_{\text{sub.1}}$, from which the core **34** begins to move from its open position to

its closed position.

(50) Next in the closing phase P2, the movable contacts **26** reach abutment against the fixed contacts **28**; the contactor **1** is then in the closed state. The core continues to move until it makes contact with the fixed portion **30** of the magnetic circuit, this corresponding to the overrun that compresses the electrical contacts.

(51) More generally, when the electrical power source comprises a plurality of electrical phases, the pads **29** associated with each of the electrical phases do not all have the same wear, or in other words do not all have the same compression. The exact time of the closure of each of the electrical phases differs depending on the electrical phase.

(52) In a last phase **64** of the closing phase P2, transitory effects have ended, and the coil current $I_{\text{sub.BOB}}$ exhibits a plateau value, equal to the coil voltage $U_{\text{sub.BOB}}$ divided by the coil resistance $R_{\text{sub.BOB}}$ and strictly higher than the stall current $I_{\text{sub.S}}$. The contactor **1** is then stably in the closed state.

(53) In the phase **64**, the coil **32** consumes electrical power, in particular via Joule heating, the power consumed by the coil **32** via Joule heating being equal to $R_{\text{sub.BOB}} \times (I_{\text{sub.BOB}})^2$. The contactor **1** remains in the closed state provided that the coil current $I_{\text{sub.BOB}}$ is kept above the stall current $I_{\text{sub.S}}$. It is therefore possible to decrease the power consumption of the coil **32** while keeping the contactor **1** stably in the closed state $I_{\text{sub.BOB}}$, by decreasing the magnitude of the coil current $I_{\text{sub.BOB}}$, provided that the coil current $I_{\text{sub.BOB}}$ remains higher than the stall current $I_{\text{sub.S}}$.

(54) This is for example achieved by varying the coil voltage $U_{\text{sub.BOB}}$, so as to decrease as much as possible the coil current $I_{\text{sub.BOB}}$ while maintaining it above the stall current $I_{\text{sub.S}}$. Such a situation corresponds to a holding phase P3.

(55) In the illustrated example, the variation in the coil voltage $U_{\text{sub.BOB}}$ is achieved by alternatively opening and closing the supply switch T2, this creating a periodic signal, chopping the coil voltage $U_{\text{sub.BOB}}$ with a square-wave profile at a frequency F3.

(56) In the holding phase P3, when the supply switch T2 is open, the release switch T1 remains closed. The command circuit **51** is then in a mode referred to as the “flyback” mode. The command circuit **51** being limited to the coil **32** connected to the flyback diode Drl and to the shunt resistor Rsh. The coil current $I_{\text{sub.BOB}}$ then decreases, the electrical energy mainly being dissipated by the resistance $R_{\text{sub.BOB}}$ of the coil **32**. The supply switch T2 is then closed before the coil current $I_{\text{sub.BOB}}$ decreases below the stall current $I_{\text{sub.S}}$, and the coil current $I_{\text{sub.BOB}}$ once again increases.

(57) As a result, the coil current $I_{\text{sub.BOB}}$ varies periodically at a frequency equal to the frequency F3 of the coil voltage $U_{\text{sub.BOB}}$, the coil current $I_{\text{sub.BOB}}$ here having a saw-tooth profile, varying between a low threshold $I_{\text{sub.1}}$, strictly higher than the stall current $I_{\text{sub.S}}$, and a high threshold $I_{\text{sub.2}}$, higher than the low threshold $I_{\text{sub.1}}$. The low threshold $I_{\text{sub.1}}$ is for example chosen to be 5% higher than the stall current $I_{\text{sub.S}}$. The high threshold $I_{\text{sub.2}}$ especially depends on the characteristics of the coil **32**, such as the coil resistance $R_{\text{sub.BOB}}$ and the coil inductance $L_{\text{sub.BOB}}$.

(58) The periodic nature of the coil current $I_{\text{sub.BOB}}$ causes mechanical vibration of the electromagnetic actuator **30**. To prevent this vibration from generating noise perceivable to the human ear, the frequency F3 is advantageously chosen to be lower than 100 Hz, or higher than 25 kHz. In the illustrated example, the frequency F3 is 100 Hz.

(59) The opening phase P4 starts when the electronic control device **40** receives in order to open, at a time $t_{\text{sub.2}}$.

(60) The electronic control device **40** commands the electromagnetic actuator **30** to open, this being achieved here by opening the supply switch T2 and opening the release switch T1, so as to make drop the voltage across the terminals of the coil **32**. The coil current $I_{\text{sub.BOB}}$ then flows through the flyback diode Drl, the coil **32**, the flyback device Dz and the shunt resistor Rsh. The control

circuit **51** is then in a mode referred to as the “release” mode, in which the coil current $I_{\text{sub.BOB}}$ decreases faster than in the flyback mode.

(61) When the coil current $I_{\text{sub.BOB}}$ drops below the stall current I_s , the return member **36** and the mechanism **290** repulse the movable contacts **24** from their closed position to their open position.

(62) Once transient induction effects have ended, the coil current $I_{\text{sub.BOB}}$ is zero, and the contactor **1** is once again stably in the open state.

(63) The electrical and electromagnetic quantities of the coil **32** will now be described.

(64) A magnetic flux ϕ is generated when the coil current $I_{\text{sub.BOB}}$ flows through the coil **32**. The value of the magnetic flux ϕ especially depends on the value of the coil current $I_{\text{sub.BOB}}$ and on the position of the movable core **34**.

(65) For example, the value of the magnetic flux ϕ is related to the values of the coil voltage $U_{\text{sub.BOB}}$ and coil current $I_{\text{sub.BOB}}$ by the following equation, denoted Math 1 below:

$$(66) \quad U_{\text{BOB}} = R_{\text{BOB}} \cdot I_{\text{BOB}} + N \frac{d\phi}{dt} \quad [\text{Math 1}]$$

in which N is the number of turns of the coil **32** and ϕ is the magnetic flux passing through each turn of the coil **32**.

(67) A position x is defined as a position of the core **34** with respect to the coil **32**. In many embodiments, the core **34** may move translationally with respect to the coil **32** along an axis of movement. The position x is then defined along this axis of movement. By convention, the position x is zero when the contactor **1** is in the open state. For each of the electrical phases of the contactor **1**, the position of the movable contacts **24** is thus related to the position x . By extension, the position x also represents a position of the electromagnetic actuator **30** or of the contactor **1**.

(68) By deriving ϕ in the equation Math 1, a general equation Math 2 describing the electromagnetic quantities in the actuator **1** is obtained:

$$(69) \quad U_{\text{BOB}} = R_{\text{BOB}} \cdot I_{\text{BOB}} + N \frac{dI_{\text{BOB}}}{dt} \frac{dx}{dt} + N \frac{dx}{dx} \frac{d\phi}{dt} + N \frac{d\phi}{di_f} \frac{di_f}{dt} \quad [\text{Math 2}]$$

in which the last term

$$(70) \quad N \frac{d\phi}{di_f} \frac{di_f}{dt}$$

contains eddy currents, which are denoted $i_{\text{sub.f}}$.

(71) In the context of the present invention, the contactor **1** has a structure that limits the generation of eddy currents $i_{\text{sub.f}}$, this allowing the last term of the equation Math 2 to be neglected.

According to some nonlimiting examples, the contactor **1** has a sheet-based laminated structure, which is produced by stacking cut sheets and which has a very low conductivity along the stacking axis of the sheets, especially because of sheet edge discontinuities. The resulting low overall conductivity is the cause of the few eddy currents generated by this type of contactor **1**.

(72) An illustration of this effect of decrease in eddy currents $i_{\text{sub.f}}$ is shown in FIG. 4, which shows a graph **66** illustrating the variation in the coil current $I_{\text{sub.BOB}}$ and in the magnetic flux ϕ in a phase of closing the contactor **1**. The absence of eddy currents $i_{\text{sub.f}}$ manifests itself in the absence of phase shift between the coil flux ϕ and the coil current $I_{\text{sub.BOB}}$, which each reach a plateau value at the same time.

(73) Since the eddy currents are neglected, the magnetic circuit has a reluctance Rel that is, on the one hand, dependent on the position x of the movable core **34** and on the coil current $I_{\text{sub.BOB}}$, and that is, on the other hand, related to the magnetic flux ϕ and to the coil current $I_{\text{sub.BOB}}$ by the following relationship $Rel(x, I_{\text{sub.BOB}}) \cdot \phi = N \cdot I_{\text{sub.BOB}}$.

(74) In other words, the magnetic flux ϕ is a function of the position x and of the coil current $I_{\text{sub.BOB}}$. The magnetic flux ϕ is expressible in the form of an analytic relationship, or indeed, to obtain high precision, by a two-dimensional response surface generated by tools for simulating the magnetic circuit of the contactor **1**.

(75) In the vast majority of cases, the surface $\phi = f(x, I_{\text{sub.BOB}})$ has a bijective character, i.e. for a

given coil current $I_{\text{sub.BOB}}$, to a given value of the position x corresponds a single value of the magnetic flux ϕ . This allows an inverse function $x=g(\phi, I_{\text{sub.BOB}})$ giving the value of the position x as a function of the flux ϕ and of the coil current $I_{\text{sub.BOB}}$ to be reconstructed.

(76) The surface $\phi=f(x, I_{\text{sub.BOB}})$, or indeed its inverse function $x=g(\phi, I_{\text{sub.BOB}})$, is stored in the memory of the electronic control device **40**, for example in the form of a data table characteristic of the electromagnetic actuator **30**, the data table defining a bijective relationship between the position x of the core **34**, the coil flux ϕ and the coil current $I_{\text{sub.BOB}}$.

(77) The magnetic flux ϕ is also given by integration with respect to time of the equation Math 1. The following equation Math 3 is then obtained:

$$(78) \quad (t) = \int \frac{U_{\text{sub.BOB}} - R_{\text{sub.BOB}} \cdot I_{\text{sub.BOB}}}{N} \cdot dt + \phi_{\text{sub.0}} \quad [\text{Math } 3]$$

in which $U_{\text{sub.BOB}}$ and $I_{\text{sub.BOB}}$ are measured, N , dt and $R_{\text{sub.BOB}}$ are known and $\phi_{\text{sub.0}}$ is an initial value of the magnetic flux ϕ .

(79) The magnetic flux ϕ may be computed using the equation Math 3 using numerical computing methods that are implemented by the electronic control device **40**.

(80) The shorter the integration time interval dt , in other words the shorter the integration stage, the smaller the computation error. The interval dt is for example proportional to the inverse of a clock frequency of the central processing unit of the electronic control device **40**. According to some examples, the clock frequency of the device **40** is 1 kHz.

(81) When the initial flux $\phi_{\text{sub.0}}$ is zero, it is possible to compute the variation in the value of the flux ϕ directly from the measurements of $U_{\text{sub.BOB}}$ and $I_{\text{sub.BOB}}$, and to determine, using the inverse function $x=g(\phi, I_{\text{sub.BOB}})$, the variation in the position x of the movable core **34**. Thus, the position of the movable core **34** in the phase of closing the contactor may be precisely determined without there being any need to install a specific sensor such as a position sensor.

(82) In certain cases, the initial flux $\phi_{\text{sub.0}}$ is not zero, for example when certain parts of the electromagnetic actuator **30** have, over time, a remnant magnetisation, or indeed when the electromagnetic actuator **30** comprises, by design, a permanent magnet. It is then necessary to determine the initial flux $\phi_{\text{sub.0}}$ in order to decrease the imprecision in the computation of the position x . An estimate $\{\text{circumflex over } (\phi)\}_{\text{sub.0}}$ of the initial flux is defined.

(83) A first method consists in determining the value of the initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub.0}}$ using the known response surface $\phi=f(x, I_{\text{sub.BOB}})$, by estimating an initial value of the position x , which is denoted estimate $\{\text{circumflex over } (x)\}_{\text{sub.0}}$. The estimate $\{\text{circumflex over } (x)\}_{\text{sub.0}}$ may for example be computed via the coil inductance $L_{\text{sub.BOB}}$, which is known, the coil current $I_{\text{sub.BOB}}$ being measured. Then, $\{\text{circumflex over } (\phi)\}_{\text{sub.0}}=f(\{\text{circumflex over } (x)\}_{\text{sub.0}}, I)$.

(84) However such a first method of determining the estimated value $\{\text{circumflex over } (x)\}_{\text{sub.0}}$ of the position is relatively approximate and especially does not allow variability in magnetic-gap dimensions to be taken into account. For example, mating of surfaces, or contamination of the surfaces in abutment in the open position, may lead to a variation in this considered to be open estimated position $\{\text{circumflex over } (x)\}_{\text{sub.0}}$, this leading to an error in the estimate of the initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub.0}}$ and therefore to an error in the reconstruction of the position x in the closing phase.

(85) A second method, referred to as the self-correction method, is based on the fact that the movable core **34** remains stationary in the open position provided that the coil current $I_{\text{sub.BOB}}$ is lower than the start current $I_{\text{sub.D}}$, i.e. between the times $t_{\text{sub.0}}$ and $t_{\text{sub.1}}$ of the phase P2.

(86) In other words, at any time t comprised between $t_{\text{sub.0}}$ and $t_{\text{sub.1}}$, provided that the coil current $I_{\text{sub.BOB}}$ is lower than the start current $I_{\text{sub.D}}$, when the magnetic flux ϕ is computed using the equation Math 3 and when the position x at the time t is deduced using the inverse function $x=g(\phi, I_{\text{sub.BOB}})$, if the computed position x does not remain constant, in other words if $x(t) \neq x(t_{\text{sub.0}})$, it means that there is an error in the estimated initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub.0}}$. The magnetic flux ϕ at the time t is then compensated for to correct this error. The

magnetic flux ϕ is corrected a plurality of times, in a plurality of successive computations and provided that the time t is comprised between $t_{\text{sub}.0}$ and $t_{\text{sub}.1}$, until the estimated initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub}.0}$ and the actual flux $\phi_{\text{sub}.0}$ converge. As a result of the self-correction method, the initial flux $\phi_{\text{sub}.0}$ is known in a precise manner.

(87) Thus, when the coil current $I_{\text{sub}.BOB}$ exceeds the start current $I_{\text{sub}.D}$ and the position x starts to change, the initial flux $\phi_{\text{sub}.0}$ is estimated in a precise manner, this initial flux value $\phi_{\text{sub}.0}$ allowing, in its turn, the position x to be computed in a precise manner.

(88) FIG. 5 shows a graph **68** illustrating an example of implementation of the self-correction method. In this graph **68**, the curve **70** shows the magnetic flux ϕ computed with self-correction, depending on the coil current $I_{\text{sub}.BOB}$ multiplied by the number of turns, the initial position x being set, and the estimate of the initial flux $\phi_{\text{sub}.0}$ being erroneous. In the example, the initial position is $\{\text{circumflex over } (x)\}_{\text{sub}.0}=0$ mm, whereas the initial flux $\phi_{\text{sub}.0}$ is the magnetic flux corresponding to a position $x=+0.02$ mm. It may be seen that there are divergences in the magnetic flux ϕ with respect to a characteristic corresponding to a set position x , leading to a series of self-corrections until the estimated flux and the real flux ϕ converge. The convergence of the estimated flux leads to a stable computed position x , which corresponds to the actual position x .

(89) Curve **72** shows the magnetic flux ϕ computed with self-correction, without error in the initial flux $\phi_{\text{sub}.0}$. In the example, $\{\text{circumflex over } (x)\}_{\text{sub}.0}=+0.02$ mm, and the initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub}.0}$ is the magnetic flux corresponding to this position $x=+0.02$ mm. The computed flux indeed follows a characteristic of set position and requires no correction.

(90) FIG. 6 shows a graph **74** illustrating the effect that the self-correction method has on the determination of the position x . A curve **76** (dashed line) shows the variation in the position x , the position x being measured directly using a sensor of the position of the movable core **34** for the purposes of comparison with the results of the method described above. It will be understood that in practice the contactor **1** is normally devoid of such a position sensor.

(91) In an initial phase **P74**, the measured position x remains constant. In a phase **P75**, after the initial phase **P74**, the core **34** moves from its closed position to its open position, and the position x varies. In a phase **P76**, after the phase **P75**, the core **34** is in its stable open position, and the position x no longer varies.

(92) A curve **78** (thick dashed line) shows the variation in the position x , computed using the method described above without self-correction. The estimated initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub}.0}$ is erroneous and is not corrected.

(93) A curve **80** shows the variation in the position x in a closing phase, computed using the method described above, with self-correction, the initial conditions being identical to those of the computation of the curve **78**. The estimated initial flux $\{\text{circumflex over } (\phi)\}_{\text{sub}.0}$ is erroneous but is corrected.

(94) In an initial phase **P74**, the position x remains constant. Curve **80** contains stair-shaped changes **81** that result from the implementation of the self-correction procedure. Moreover, curve **80** joins curve **76** despite the uncertainty in the initial flux $\phi_{\text{sub}.0}$. As a result, in phases **P75** and **P76**, curve **80** corresponds better to curve **76** than curve **78**, this demonstrating the positive effect of the self-correction method. It is not necessary to install a position sensor to reliably determine the variation in the position x as a function of time in the closing phase **P2**. The self-correction method is advantageously implemented in all cases, whether the electromagnetic actuator **30** comprises a permanent magnet or not.

(95) The data on the position of the movable core **34** that are thus generated allow various conclusions to be drawn as to the state of the contactor **1**.

(96) For example, reconstruction of the position $x=f(t)$ allows the conformity of the travel of the movable core **34** and the final state of the actuator **1** to be verified, in order to detect potential defects of mechanical nature.

(97) Through a closing time, equal to a time difference between the order to close and the end of

mechanical travel of the actuator **1**, it is possible to verify the absence of friction or blockages of mechanical origin.

(98) By differentiating the position $x=f(t)$ with respect to time, a speed of movement of the movable core **34** is computed. In the closing phase **P2**, a minimum speed in the overrun that leads to the compression of the moveable contacts **24** is an advantageous observer of the ability of the contactor **1** to close stably. Too low a minimum speed is an indication of a lack of magnetic force or of excessively high friction within the closing device **30**.

(99) An example of operation of a method for estimating properties of the contactor **1** according to one preferred embodiment will now be described with reference to FIG. 7. However, as a variant, the steps of this method could be executed in a different order. Some steps could be omitted. The described example does not prevent, in other embodiments, other steps from being implemented conjointly and/or sequentially with the described steps.

(100) In a step **100**, the method is initialized while the contactor **1** is initially in a stable open state, in the initial phase **P1** for example. The electronic control device **40** is waiting for an order to close the contactor **1**.

(101) Next, in a step **102**, the electronic control device **40** receives an order to close the contactor **1**. The time $t_{sub.0}$ of reception of the order to close is the start of the closing phase **P2**.

(102) Next, in a step **104**, the electronic control device **40** commands the actuator **30** to close. The command to close here consists in applying the command coil voltage $U_{sub.BOB}$ across the terminals of the coil **32** by closing the switches **T1** and **T2**.

(103) In the example illustrated in FIG. 3, the order to close is received in the particular case where the coil current $I_{sub.BOB}$ is non-zero, following a transient peak **62** related to a testing operation. In some examples, the coil current $I_{sub.BOB}$ is zero when the order to close is received. In all cases, the electromagnetic actuator **30** is in its stable open position when the order to close is received and when the electronic control device **40** commands the electromagnetic actuator **30** to close, i.e. the coil current $I_{sub.BOB}$ is strictly lower than the start current $I_{sub.D}$.

(104) Next, in a step **106**, the electronic control device **40** measures and stores the coil current $I_{sub.BOB}$ and coil voltage $U_{sub.BOB}$, the measurements continuing for example until the electromagnetic actuator **30** stably reaches its closed position, or indeed for a length of time which is stored in the electronic control device **40**, this length of time being equal for example to 5 periods of a 50 Hz supply current, i.e. to 100 ms.

(105) The fact that the electromagnetic actuator **30** is stably in the closed position may be detected in various ways. For example, by testing the position x of the core **34**, or else when the coil current $I_{sub.BOB}$ reaches a plateau value, substantially equal to $U_{sub.BOB}/R_{sub.BOB}$, in the phase **64**.

(106) Next, the method comprises a step **112** of computing and storing the magnetic flux ϕ flowing through the coil **32**.

(107) Step **112** comprises a substep **110** in which the magnetic flux ϕ is computed by integration of the stored values of the coil current $I_{sub.BOB}$ and coil voltage $U_{sub.BOB}$ and of values, stored beforehand in the electronic control device **40**, of resistance $R_{sub.BOB}$ and inductance $L_{sub.BOB}$ of the coil **32**.

(108) Step **112** advantageously comprises an initial self-correction substep **108**, after the substep **110**, the self-correction substep consisting in, provided that the coil current $I_{sub.BOB}$ is lower than the start current $I_{sub.D}$ and that the core **34** is kept in open position by the return member **36** of the actuator **1**, evaluating and storing an initial value of the magnetic flux $\phi_{sub.0}$, the computation by integration of the magnetic flux ϕ in the substep **110** taking into account the value of the initial flux $\phi_{sub.0}$ computed in the self-correction substep **108**.

(109) According to some examples, the self-correction substep **108** lasts from the time $t_{sub.0}$ to the time $t_{sub.1}$, whereas substep **110** starts at the time $t_{sub.1}$. In other words, when the coil current $I_{sub.BOB}$ increases above the start current $I_{sub.D}$, the computation by integration of the magnetic flux ϕ in the computing substep **110** takes into account the value of the initial flux

ϕ .sub.0.

(110) Preferably, in step **104**, the electronic control device **40** commands the electromagnetic actuator **30** to close when the coil current I.sub.BOB is sufficiently low, so that the self-correcting substep **108** lasts long enough for the computation of the initial flux ϕ .sub.0 to converge. Thus, in the step **104** of commanding the actuator **30** to close, the electronic control device **40** commands the electromagnetic actuator **30** to close when the coil current I.sub.BOB reaches 50% of the start current I.sub.D, preferably 20%, and more preferably 10%, of the start current I.sub.D.

(111) Next, the method comprises a step **114** consisting in evaluating and storing the position x of the contactor **1** on the one hand on the basis of the stored values of the magnetic flux ϕ and of the coil current I.sub.BOB and, on the other hand, depending on a data table characteristic of the coil **32**, the data table being stored beforehand in the electronic control device **40** and defining a bijective relationship between the position x of the core **34**, the magnetic flux ϕ and the coil current I.sub.BOB.

(112) In a step **116** of the method, a profile of the speed v of movement of the contactor **1** and a profile of the acceleration α of the contactor **1** are computed by differentiation with respect to time of the values of position x stored in step **114**. The profiles of speed v and acceleration α are stored in the electronic control device **40**. By extension, the profiles of the speed v and acceleration α of the contactor **1** are also profiles of the speed v and acceleration α of the electromagnetic actuator **30** or of the movable core **34**.

(113) The diagnosing method according to the invention makes it possible to precisely determine the profile of the position x of the movable core **34** as a function of time, and profiles of velocity v and acceleration α , which profiles are derived from the profile of the position x . This information may be combined in many ways with a view to monitoring wear of specific components of the contactor **1**, for example, the contact pads **29**, or indeed the mechanism of the electromagnetic actuator **30**.

(114) According to a first example, in a step **118**, a travel of the core **34**, which travel is equal to the difference between the positions of the core **34** in its stable closed and open position, is evaluated on the basis of the profile of the position x . The travel of the core **34** is then compared to a reference value, which is stored beforehand in the electronic control device **40**, so as to detect potential defects of mechanical nature. The reference value of the travel of the core **34** is for example equal to the travel of the core **34** evaluated when the contactor **1** is first installed, i.e., when the contactor **1** is new.

(115) According to another example, a closing time of the electromagnetic actuator **30** is evaluated on the basis of the profile of the position x , which profile is stored in the electronic control device **40**, the closing time being equal to a time difference between the time at which, in step **104**, the electronic control device **40** commands the electromagnetic actuator **30** to close and the time at which the actuator **30** reaches its stable closed position. The closing time of the actuator **30** is then compared to a reference value, stored beforehand in the electronic control device **40**, so as to detect the absence of friction or blockages of mechanical origin. The reference value of the travel of the core **34** is for example a specification of the manufacturer of the contactor **1**, or indeed otherwise equal to the closing time measured when the contactor **1** is first installed, i.e. when the contactor **1** is new.

(116) According to yet another example, in a step **120**, a minimum speed of compression of the core **34** is measured, the minimum speed of compression being equal to a minimum of the speed v of the core **34** in a terminal segment of the movement of the core **34** in the closing phase. For example, the terminal segment of the movement of the core **34** is located between the position x corresponding to the time at which the contacts **24** and **28** touch and the position in which the actuator **30** is closed and stable.

(117) The minimum speed of compression is then compared to a reference value, stored beforehand in the electronic control device **40**.

(118) The minimum speed of compression of the core **34** is an advantageous indicator of the ability of the contactor **1** to close stably. Too low a minimum compression speed is a sign that the magnetic force of the electromagnetic actuator **30** is too low, or indeed that the mechanism is generating too much friction. Thus, if the minimum compression speed of the core **34** is lower than the reference value, the electronic control device **40** sends an alarm signal to an operator.

(119) According to other examples, when the contactor **1** is connected to a multiphase power source, and in particular to a three-phase power source, the position profile $x=f(t)$ is combined with other measurements taken on each of the poles of the contactor **1**, and for example with measurements of current that are carried out by means of sensors **46**. The wear status of the contact pads **29** associated with each of the poles being a priori different, in a phase P2 of closing the contactor **1**, the contact pads **29** do not make electrical contact at the same moment. There is therefore a desynchronization of the poles that is related to the dispersion or to the wear of the parts of the contactor **1**.

(120) For example, as described in patent application WO-03/054895-A1, current measurements are carried out for each of the poles of the contactor **1**. The appearance of current in each of the poles allows the times of closure of the contacts **24** associated with each pole to be detected. By combining this information with the position profile $x=f(t)$ of the core **34**, which is obtained by virtue of the present invention and stored in the electronic control device **40**, it is possible to precisely evaluate a position of the contacts **24** when they make contact with the corresponding fixed contacts **28**, and to deduce therefrom a difference in compression associated with the contacts of each of the poles of the contactor **1**.

(121) A final position $\{\text{circumflex over (x)}\}_{\text{sub.end}}$ is defined as being a stable position of the core **34** in its closed position. For each contact **24** in question, the compression of this contact is equal to the difference between the final position $\{\text{circumflex over (x)}\}_{\text{sub.end}}$ and the position of the core **34** at the precise time of closure of the contactor **24** in question.

(122) In the particular example of a contactor **1** comprising three poles, each associated with one phase of a three-phase power source, the contacts **29** associated with each of the poles close successively. On closure of the first movable contacts **24**, which are also called the first contacts **24A**, no current is yet able to flow.

(123) A position $\{\text{circumflex over (x)}\}_{\text{sub.i1}}$ is defined as being the position of the core **34** when the first contacts **24A** make contact with the corresponding fixed contacts **28**. A difference between the position $\{\text{circumflex over (x)}\}_{\text{sub.i1}}$ and the final position $\{\text{circumflex over (x)}\}_{\text{sub.end}}$ defines an estimation of the compression of the first contacts **24A**. In other words, $E_{\text{cr.sub.1}} = \{\text{circumflex over (x)}\}_{\text{sub.i1}} - \{\text{circumflex over (x)}\}_{\text{sub.end}}$.

(124) On closure of the second movable contacts **24**, which are also called the second contacts **24B**, a current $i_{\text{sub.2}}$ appears and flows between the first contacts **24A** and second contacts **24B**. The current $i_{\text{sub.2}}$ is detected by the current sensor **46**. The position x of the core **34** at this time, which is called the position $\{\text{circumflex over (x)}\}_{\text{sub.i2}}$, is therefore the position of the core **34** when the second contacts **24B** make contact with the corresponding fixed contacts **28**.

(125) A difference between the position $\{\text{circumflex over (x)}\}_{\text{sub.i2}}$ of the second contacts **24B** and the final position $\{\text{circumflex over (x)}\}_{\text{sub.end}}$ defines an estimation of the compression of the second contacts **24B**, which compression is denoted $E_{\text{cr.sub.2}}$. In other words, $E_{\text{cr.sub.2}} = \{\text{circumflex over (x)}\}_{\text{sub.i2}} - \{\text{circumflex over (x)}\}_{\text{sub.end}}$.

(126) On closure of the third movable contacts **24**, which are also called the third contacts **24C**, a current $i_{\text{sub.3}}$ appears and flows between the three poles of the contactor **1**. The position x of the core **34** at this time, which is called the position $\{\text{circumflex over (x)}\}_{\text{sub.i3}}$, is therefore the position of the core **34** when the third contacts **24B** make contact with the corresponding fixed contacts **28**.

(127) The difference between the position $\{\text{circumflex over (x)}\}_{\text{sub.i3}}$ of the third contacts **24C** and the final position $\{\text{circumflex over (x)}\}_{\text{sub.end}}$ defines an estimation of the compression of

the third contacts, which compression is denoted $E_{cr,3}$. In other words $E_{cr,3} = \{\text{circumflex over (x)}\}_{i3} - \{\text{circumflex over (x)}\}_{end}$.

(128) In the case of supply of power to a three-phase electrical load **23** such as a motor, only the compressions $E_{cr,2}$ and $E_{cr,3}$ are measured, because only the times of appearance of the currents i_2 and i_3 are detected. In contrast, the compression $E_{cr,1}$ of the first contacts **24A** cannot be measured because no current appears when the first contacts **24A** make contact with the corresponding fixed contacts **28**—it is only possible to deduce that the compression $E_{cr,1}$ of the first contacts **24A** is greater than the compression $E_{cr,2}$ of the second contacts **24B**.

(129) According to other examples, by combining the information relative to the times of appearance of the currents on each of the poles with the profile of the speed v of the core **34**, which is obtained by virtue of the present invention and stored in the electronic control device **40**, it is possible to precisely evaluate a speed of the contacts **24** when they make contact with the corresponding fixed contacts **28**, and to deduce therefrom a difference in compression associated with the contacts of each of the poles of the contactor **1**.

(130) In the illustrated example, the electronic control device **40** performs both the function of controlling the electromagnetic actuator **30** and the function of implementing the diagnosing method according to the invention.

(131) As a variant (not shown) the diagnosing method is implemented by a specific electronic device, which is added to an existing contactor **1** during a maintenance operation. Such a specific device requires, to implement the diagnosing method according to the invention, only measurements of the coil current I_{BOB} and coil voltage U_{BOB} and optionally measurements of the power currents flowing through the upstream line **20**.

(132) According to another variant, when a contactor **1** comprises an electronic control device **40** that is not configured to implement the diagnosing method according to the invention, if sensors of coil current and coil voltage are present, it is possible to configure the electronic control device **40** of the contactor **1** so that it is able to implement the diagnosing method according to the invention, for example by physically replacing the electronic control device **40** or indeed, where appropriate, by changing the computational code of the electronic control device **40**.

(133) The embodiments and the variants mentioned above may be combined with one another to create new embodiments of the invention.

Claims

1. A method for diagnosing an operating state of a switching apparatus, the switching apparatus being configured to be coupled to an electrical conductor and comprising: separable contacts that are associated with the electrical conductor and that are driven by an electromagnetic actuator comprising a coil connected to an electronic control device that is configured to apply a coil command voltage across the terminals of the coil, and sensors that are configured to measure a magnitude of a coil current flowing through the coil and the coil voltage, the method comprising diagnosing an operating state of the switching apparatus which comprises a contactor, the diagnosing including: a) receiving an order to close the switching apparatus, the switching apparatus initially being in an open state, the order to close being received by the electronic control device; b) after the order to close has been received, commanding the electromagnetic actuator to close, by means of the electronic control device; c) as the switching apparatus switches to the closed state, measuring values of the coil voltage and coil current using the sensors and storing the values of the coil voltage and coil current; d) computing and storing values of a magnetic flux passing through the coil based on the stored values of the coil current and coil voltage and on values of the resistance and inductance of the coil that are stored beforehand in the electronic control device; and e) on the basis of the stored values of the magnetic flux and coil current, evaluating and storing positions of a core of the electromagnetic actuator depending on a data table

characteristic of the electromagnetic actuator, the data table being stored beforehand in the electronic control device and defining a bijective relationship between the position of the core, the magnetic flux and the coil current, wherein computing values of the magnetic flux comprises an initial substep, referred to as the self-correction substep, followed by a computing substep, the self-correction substep comprising applying a current to the coil lower than a current referred to as the “start” current, keeping the movable core in the open position by a return member of the electromagnetic actuator, and evaluating and storing an initial value of the magnetic flux, said value being referred to as the “initial flux”, so that, when the coil current increases above the start current, the computation of the magnetic flux takes into account the value of the initial flux.

2. The method according to claim 1, wherein, in commanding the electromagnetic actuator to close, the electronic control device commands the electromagnetic actuator to close when the coil current reaches a given percentage of the start current that is 50% or below of the value of the start current so that the self-correction substep lasts long enough for the computation of the initial flux to converge.
3. The method according to claim 1, further comprising computing, from a profile of the position stored in the electronic control device, a closing time of the electromagnetic actuator, the closing time being equal to a time difference between the time at which the electronic control device commands the electromagnetic actuator to close and the time at which the electromagnetic actuator reaches its stable closed position.
4. The method according to claim 1 further comprising computing, by derivation with respect to time of a profile of the position stored in the electronic control device, a profile of the speed of movement of the switching apparatus and a profile of the acceleration of the switching apparatus, and storing the speed and acceleration profiles in the electronic control device.
5. The method according to claim 4, further comprising computing, from the profile of the speed of the switching apparatus, a minimum compression speed of the core, the minimum compression speed being equal to a minimum of the speed of the core in a terminal segment of the movement of the core in a closing phase, the closing phase being an interval over which the switching apparatus switches to the closed state.
6. The method according to claim 4 further comprising detecting precise times of closure of each contact associated with each pole of the switching apparatus by means of current sensors, and combining the profile of the speed stored in the electronic control device, and the precise times of closure of each contact, in order to evaluate a closing speed of each contact.
7. The method according to the claim 6, wherein a compression associated with the movable contacts of each pole is evaluated by combining the profile of the position of the core at the precise times of closure of each contact, the compression of each movable contact being equal to the difference between a final position of the core in its stable closed position and the position of the core at the precise time of closure of the contact in question.
8. The method according to claim 1, wherein the magnetic flux is computed by integration of the stored values of the coil current and coil voltage and of a value of the resistance of the coil that is stored beforehand in the electronic control device, and by using a value related to the inductance of the coil stored beforehand in the electronic control device.
9. The method according to claim 1, wherein the positions are evaluated and stored in relation to time to provide a profile of the position of the core of the electromagnetic actuator as a function of time, and the profile of the position of the core or information derived from the profile is evaluated to monitor wear of the switching apparatus or a component thereof.
10. The method according to claim 1, wherein a travel of the core of the electromagnetic actuator of the contactor between the open state and the closed state is evaluated on the basis of a profile of the positions of the core; and the travel of the core of the electromagnetic actuator is compared to a reference value, which is stored beforehand in the electronic control device, so as to detect for a potential defect of a mechanical nature of the contactor.

11. The method according to claim 1, wherein computing values of the magnetic flux compensates for variability of magnetic-gap dimensions of the contactor.

12. The method according to claim 11, wherein computing values of the magnetic flux includes: compensating an estimated initial magnetic flux to correct for variability of magnetic-gap dimensions of the contactor.

13. An electrical switching apparatus configured to be coupled to an electrical conductor and comprising: separable contacts that are moved between an open position and a closed position by an electromagnetic actuator comprising a coil and a movable core that is attached to the separable contacts, the switching apparatus having a structure that limits the generation of eddy currents; a command circuit for controlling the voltage across the terminals of the coil, said voltage being referred to as the “coil voltage”, the command circuit comprising a device referred to as the “release” device, which may be selectively activated in order to make drop the electrical current flowing through the coil, which is referred to as the “coil current”, the coil voltage and the release device being activated or deactivated depending on the states of the command circuit; sensors for measuring the coil current and coil voltage; and an electronic control device that is configured to receive orders to open and close the switching apparatus, to receive the values of the measurements of coil current and coil voltage, and to control the states of the command circuit; wherein the switching apparatus comprises a contactor and is configured to diagnose an operating state of the switching apparatus, by implementing a diagnosing method comprising: a) receiving an order to close; b) commanding the electromagnetic actuator to close; c) measuring values of the coil voltage and coil current using the sensors and storing the values of the coil voltage and coil current; d) computing and storing values of a magnetic flux passing through the coil based on the stored values of the coil current and coil voltage and on values, which are stored beforehand in the electronic control device, of the resistance and inductance of the coil; and e) on the basis of the stored values of the magnetic flux and coil current, evaluating and storing positions of the core depending on a data table characteristic of the electromagnetic actuator, the data table defining a bijective relationship between the position of the core, the magnetic flux and the coil current, wherein computing values of the magnetic flux comprises an initial substep, referred to as the self-correction substep, followed by a computing substep, the self-correction substep comprising applying a current to the coil lower than a current referred to as the “start” current, keeping the movable core in the open position by a return member of the electromagnetic actuator, and evaluating and storing an initial value of the magnetic flux, said value being referred to as the “initial flux”, so that, when the coil current increases above the start current, the computation of the magnetic flux takes into account the value of the initial flux.

14. The electrical switching apparatus according to claim 13, wherein the switching apparatus comprises current sensors that are configured to measure a magnitude of the electrical current flowing through each of a plurality of phases of the electrical conductor, the switching apparatus being configured to implement a diagnosing method comprising detecting a precise time of closure of the movable contacts associated with each phase, and combining with the precise times of closure of each of the movable contacts the profile of the position of the core or the profile of the speed of the core, in order to evaluate a compression or a closing speed of each of the movable contacts.

15. The electrical switching apparatus according to claim 13, wherein the magnetic flux is computed by integration of the stored values of the coil current and coil voltage and of a value of the resistance of the coil that is stored beforehand in the electronic control device, and by using a value related to the inductance of the coil stored beforehand in the electronic control device.

16. The electrical switching apparatus according to claim 13, wherein the positions are evaluated and stored in relation to time to provide a profile of the position of the core of the electromagnetic actuator as a function of time, and the profile of the position of the core or information derived from the profile is evaluated to monitor wear of the switching apparatus or a component thereof.

17. The electrical switching apparatus according to claim 13, wherein a travel of the core of the electromagnetic actuator of the contactor between an open state and a closed state is evaluated on the basis of a profile of the positions of the core; and the travel of the core of the electromagnetic actuator is compared to a reference value, which is stored beforehand in the electronic control device, so as to detect for a potential defect of a mechanical nature of the contactor.

18. The electrical switching apparatus according to claim 13, wherein computing values of the magnetic flux compensates for variability of magnetic-gap dimensions of the contactor.

19. The electrical switching apparatus according to claim 18, wherein computing values of the magnetic flux includes: compensating an estimated initial magnetic flux to correct for variability of magnetic-gap dimensions of the contactor.
